EXPERIMENT - 10

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AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

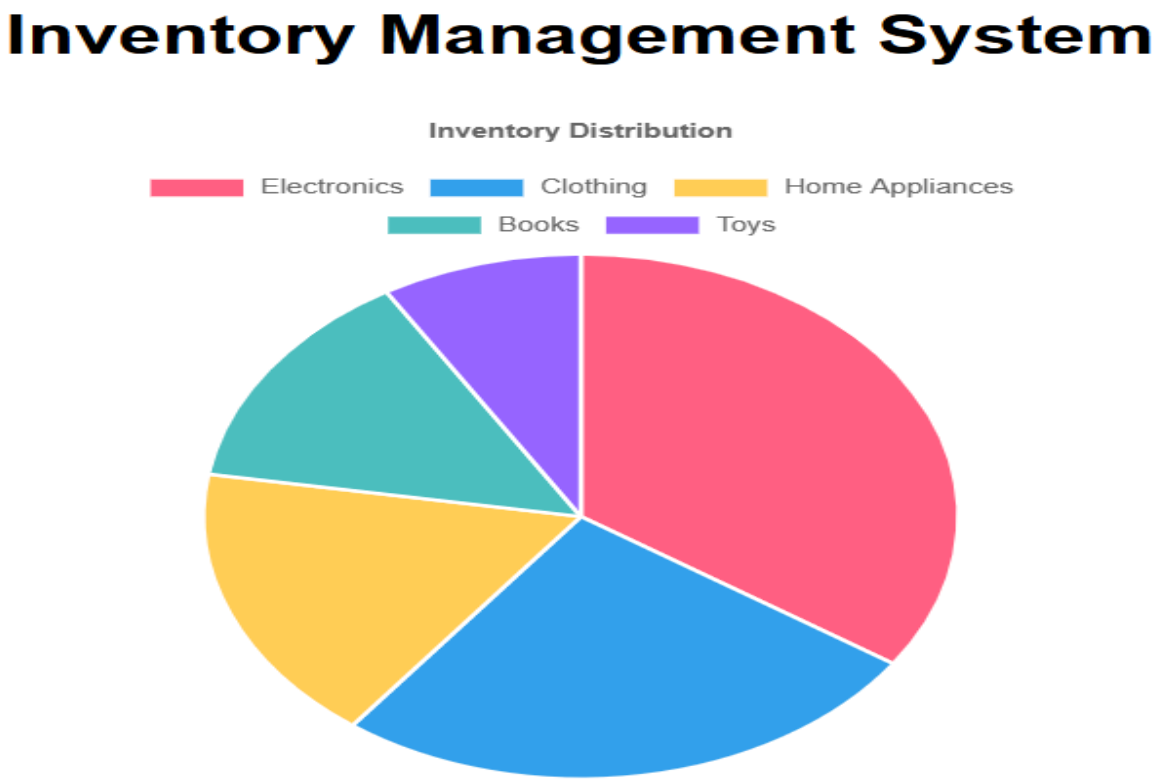
Step 1:

Set Up Your HTML File :

```
<!DOCTYPE html>

<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Inventory Management Visualization</title>

```
<style>
   body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin: 50px;
   }
   canvas {
      margin: 20px auto;
   }
</style>
```

</head>

<body>

```

```
<h1>Inventory Management System</h1>

<!-- Pie Chart -->
<canvas id="pieChart" width="400" height="400"></canvas>

<!-- Bar Chart -->
<canvas id="barChart" width="400" height="400"></canvas>

<!-- Chart.js Library -->
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<!-- Your JS file -->
<script src="script.js"></script>

</body>
</html>
```

OUTPUT:



Step 2:

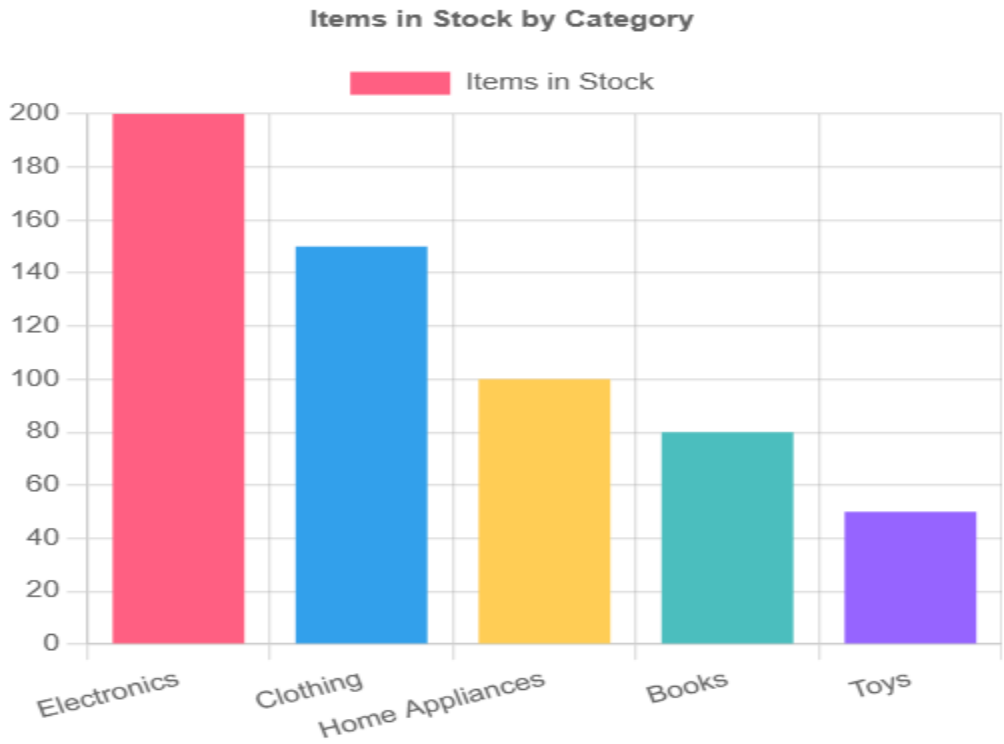
Create the JavaScript File for Charts Next, create a JavaScript file (script.js) to handle the data visualization logic.

```
// Inventory Data
const inventoryData = {
labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
datasets: [{
label: 'Items in Stock',
data: [200, 150, 100, 80, 50],
backgroundColor: [
'#FF6384',
'#36A2EB',
'#FFCE56',
'#4BC0C0',
```

```
'#9966FF'
]
}]
};

// PIE CHART
const ctxPie = document.getElementById('pieChart').getContext('2d');
new Chart(ctxPie, {
type: 'pie',
data: inventoryData,
options: {
responsive: true,
plugins: {
title: {
display: true,
text: 'Inventory Distribution'
}
}
}
});

// BAR CHART
const ctxBar = document.getElementById('barChart').getContext('2d');
new Chart(ctxBar, {
type: 'bar',
data: inventoryData,
options: {
responsive: true,
plugins: {
title: {
display: true,
text: 'Items in Stock by Category'
}
},
scales: {
y: {
beginAtZero: true
}
}
}
```

});

OUTPUT:



EXPERIMENT - 10 B

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1:

Set Up Your HTML File

index.html > ...

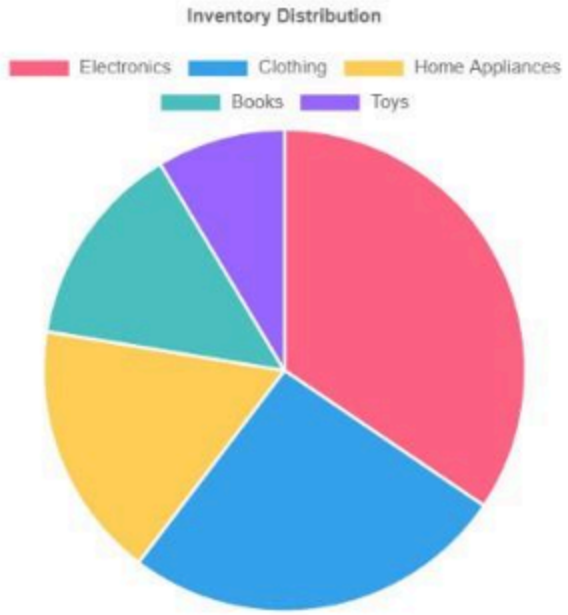
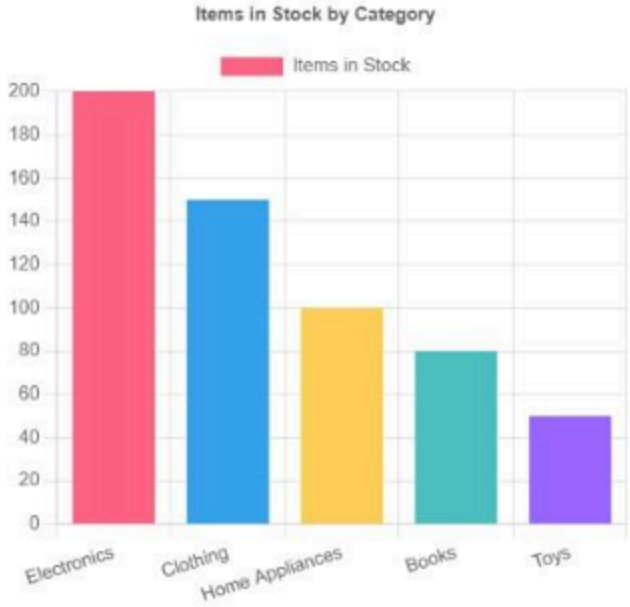
```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Inventory Management Visualization</title>
<style>
body {
font-family: Arial, sans-serif;
text-align: center;
margin: 50px;
}
canvas {
margin: 20px auto;
}
</style>
</head>
<body>
<h1>Inventory Management System</h1>
<canvas id="pieChart" width="400" height="400"></canvas>
<canvas id="barChart" width="400" height="400"></canvas>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>
</body>
</html>
```

Step 2:

Create the JavaScript File for Charts Next, create a JavaScript file (script.js) to handle the data visualization logic.

JS script.js > ...

```javascript
// Data for inventory
const inventoryData = {
    labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
    datasets: [{
      label: 'Items in Stock',
      data: [200, 150, 100, 80, 50],
      backgroundColor: [
        '#FF6384', '#36A2EB', '#FFCE56', '#4BC0C0', '#9966FF'
      ]
    }]
  };

  // PIE Chart
  const ctxPie = document.getElementById('pieChart').getContext('2d');
  new Chart(ctxPie, {
    type: 'pie',
    data: inventoryData,
    options: {
      responsive: true,
      plugins: {
        title: {
          display: true,
          text: 'Inventory Distribution'
        }
      }
    }
  });

  // BAR Chart
  const ctxBar = document.getElementById('barChart').getContext('2d');
  new Chart(ctxBar, {
    type: 'bar',
    data: inventoryData,
    options: {
      responsive: true,
      plugins: {
        title: {
          display: true,
          text: 'Items in Stock by Category'
        }
      },
      scales: {
        y: {
          beginAtZero: true
        }
      }
    }
  });
```

OUTPUT:

Inventory Management System